

Guidance Document on *Stenotrophomonas maltophilia*

Version 2, November 2024

The organism

Stenotrophomonas maltophilia is a ubiquitous environmental organism. In patients, it is most often associated with colonisation, but is an occasional cause of infection, particularly in immunocompromised patients and patients with cystic fibrosis [1-3]. Most commonly *S. maltophilia* is recovered from lower respiratory tract samples, in patients who are colonised or infected.

Antimicrobial resistance

This microorganism exhibits mechanisms of resistance to most antimicrobials. It is influenced by two inducible chromosomal genes, L1 (a class B3 metallo-beta-lactamase) and L2 (a serine beta-lactamase with cephalosporinase hydrolysis) which impact the activity of all beta-lactams including carbapenems and aztreonam. In addition, *S. maltophilia* features expected resistance to aminoglycosides due to the production of two proteases (ClpA and HtpX), aminoglycoside modifying enzymes such as aminoglycoside-phosphotransferase (APH-3'-IIc) or acetyltransferase (AAC-6'-Iz), and several RND efflux pumps such as SmeYZ [2]. Other efflux pumps might also compromise the activity of the old tetracyclines. The presence of the chromosomally-encoded *Smqnr* gene further diminishes the activity of fluoroquinolones. Moreover, acquired genes may be present, conferring resistance to a wide range of agents, including trimethoprim-sulfamethoxazole.

Antimicrobial activity

Epidemiological cut-off values are available for several antimicrobials (found at <https://mic.eu-cast.org>) However, the testing reliability of some of these antimicrobials can be questioned because determining MIC values for *S. maltophilia* is fraught with difficulties and variability between methods is greater than for most other organisms, as observed in the widths of the confidence intervals for ECOFFs. Sufficient data are available to establish ECOFFs or tentative ECOFFs (TECOFFs) for the most frequently used and recommended agents for treatment.

In vitro and animal model pharmacodynamic studies

There have been two recent *in vitro* time-kill studies examining the activity of trimethoprim-sulfamethoxazole, showing that at best it is bacteriostatic [4,5]. These results have been confirmed in an *in vitro* dynamic model [6].

A time-kill study examined the efficacy of minocycline, tigecycline, levofloxacin and ceftazidime [4]. In general, each of these agents showed little or no activity at concentrations close to their mean free area under the plasma concentration-time curve (AUC) at steady state in healthy human populations calculated using literature data.

Several agents with potential activity against *S. maltophilia* have been identified in the mouse thigh model, including cefiderocol, levofloxacin, and intravenous minocycline [7-10]. Due to high natural thymidine levels in mice, this model is not suitable for testing trimethoprim-sulfamethoxazole, and papers reporting PK-PD based on murine models are not reliable.

Barrasa et al. accessed data from a range of studies (references 6-14) to develop estimates of PK/PD cut-offs for 6 agents frequently considered as treatment options for *S. maltophilia* infections [15]. Identified PK/PD indices and target values are listed in Tables 1 and 2. Monte Carlo simulations were conducted and displayed graphically. In all cases, except that for cefiderocol, the observed PK/PD cut-offs were lower than the ECOFFs.

Table 1

PK/PD indices and targets determined from *in vitro* and animal studies of *S. maltophilia*

Agent [Ref]	Index	Target
Trimethoprim-sulfamethoxazole [6]	$fAUC_{24}/MIC$	67.4
Levofloxacin [7,9]	$fAUC_{24}/MIC$	62
Minocycline [7,10]	$fAUC_{24}/MIC$	8.75
Tigecycline [7]	$fAUC_{24}/MIC$	0.9

Table 2

PK/PD indices and targets determined from *in vitro* and clinical studies of other species

Agent [Ref]	Species studied	Index	Target
Cefiderocol [11]*	Multiple clinical studies	$f\%T > MIC$	75%
Aztreonam-avibactam [12-14]**	<i>E. coli in vitro</i> PK/PD model	$f\%T > MIC$ / $f\%T > 2.5$ mg/L	60% / 50%

* There were 6 cases of *S. maltophilia* infection in the clinical studies that were analysed; most cases were multidrug-resistant gram-negative infections.

** Beta-lactam/beta-lactamase inhibitor combination

Clinical studies

There has been a wide range of clinical studies of *S. maltophilia* infections over the years. These have been subject to a range of meta-analyses comparing individual agents/classes and mono-versus combination therapy. These studies are summarised in Table 3 below.

Table 3

Recent analyses of therapeutic agents for *Stenotrophomonas* infections

First author Year [Reference]	Comparison	Type of analysis	Infection types	Outcome measured and findings	Conclusion
Ko 2019 [16]	Fluoroquinolones (mainly ciprofloxacin (CIP) and levofloxacin (LEV) vs Trimethoprim-sulfamethoxazole (TRS)	Meta-analysis 14 studies included	All	<u>30-day mortality (n/N)</u> CIP 19/114 (17%) LEV 52/187 (28%) TRS 111/332 (33.4) <u>Odds ratio:</u> FQs vs TRS 0.68 (0.53-0.97)	Fluoroquinolones superior to trimethoprim-sulfamethoxazole in terms of 30-day mortality
Prawang 2022 [17]	Monotherapy (MONO) vs combination (COMB)	Meta-analysis of 4 studies included	Bacteraemia (2) and pneumonia (2)	<u>Mortality (various time points)</u> MONO (n=274) COMB (n=237) <u>Hazard ratio (favours MONO)</u> 1.42 (1.04-1.94)	Monotherapy superior to combination therapy
Maraolo 2023 [18]	Fluoroquinolones (FQ) vs Trimethoprim-sulfamethoxazole	Meta-analysis 24 studies included	All	<u>Monotherapy</u> <u>7 to 30-day mortality (n/N)</u> FQ 143/1103 (13%) TRS 247/1304 (19%) <u>Odds ratio:</u> FQs vs TRS 0.62 (0.39-0.99)	Fluoroquinolones superior to trimethoprim-sulfamethoxazole in terms of 7 to 30-day all-cause mortality
Hevia 2023 [19]	Minocycline (MIN) vs Trimethoprim-sulfamethoxazole (as initial monotherapy)	Multicentre retrospective	Non-urinary	<u>Clinical failure (n/N):</u> MIN 16/82 (19%) TRS 4/11 (36%) p = 0.242	No difference between minocycline and trimethoprim-sulfamethoxazole in terms of clinical failure rates

First author Year [Reference]	Comparison	Type of analysis	Infection types	Outcome measured and findings	Conclusion
Alhayani 2023 [20]	Doxycycline (DOX) or Minocycline (MIN) vs Trimethoprim-sulfamethoxazole	2-centre retrospective	Pneumonia	<u>Clinical success (n/N)</u> DOX/MIN 6/21 (29%) TRS 15/59 (25%) p = 0.994	No difference between doxycycline/minocycline and trimethoprim-sulfamethoxazole in terms of clinical failure rates
Almangour 2024 [21]	Trimethoprim-sulfamethoxazole versus levofloxacin	4-centre retrospective	All	<u>Clinical success (n/N)</u> TRS 173/316 (55%) LEV 35/55 (64%) Odds ratio 0.70 (0.37-1.31) <u>30-day mortality</u> TRS 88/316 (28%) LEV 14/55 (25%) Odds ratio 1.13 (0.59-2.18)	No difference between levofloxacin and trimethoprim-sulfamethoxazole in terms of clinical failure or 30-day mortality rates
Almangour 2024 [22]	Monotherapy (MONO) vs combination (COMB)	4-centre retrospective	All	<u>Clinical success (n/N)</u> MONO 181/330 (55%) COMB 50/77 (65%) Adj Odds ratio 0.72 (0.04-1.31) <u>30-day mortality</u> MONO 91/330 (28%) COMB 25/77 (32%) Adj. Odds ratio 0.45 (0.22-0.90)	No difference between monotherapy and combination therapy in terms of clinical failure rates but monotherapy superior to combination therapy in terms of 30-day mortality

Another recent study examined the risk factors for mortality from *S. maltophilia* and suggested that trimethoprim-sulfamethoxazole was similar to fluoroquinolones (mainly levofloxacin and moxifloxacin) in terms of all-cause mortality [23]. All these analyses are based on retrospective observational studies which are always subject to many potential biases. There is currently one prospective study listed on clinicaltrials.org: Efficacy of combination therapy with minocycline for treatment of *Stenotrophomonas maltophilia* infections. Results have yet to be posted.

Treatment uncertainties

Despite the findings of the most recent *in vitro* and animal model pharmacodynamic studies, and a range of meta-analyses on retrospective data, several uncertainties remain:

- Why are clinical failure and mortality rates so high? Is this a reflection of the severity or type of comorbidities?
- Does the lack of bactericidal effect and the possible inferiority of trimethoprim-sulfamethoxazole compared to fluoroquinolones suggest that the former should be replaced as the first choice for therapy?
 - If so, which agent(s)?
- Should the estimated PK/PD cut-offs being lower than the ECOFFs for all agents except cefiderocol (and maybe minocycline) mean that future studies should focus on this agent?
 - When might more clinical data become available on the efficacy of cefiderocol?
- Would a combination of aztreonam with a beta-lactamase inhibitor such as avibactam have clinical utility?
 - The PK/PD estimates provided by Barrasa et al. suggest that it would at least be unsuitable in monotherapy
- Are combination regimens truly inferior (or at least not superior) to monotherapy?
 - The studies only compare combination therapy to monotherapy and do not address the question whether some combinations may be better than others and potentially superior to monotherapy. It is also likely that sicker patients are more likely to receive combination therapy in observational trials.

Current Treatment Choices

The most frequently recommended agent is trimethoprim-sulfamethoxazole and this is the only agent for which EUCAST breakpoints were available in 2024: susceptible ≤ 0.001 mg/L; resistant > 4 mg/L, emphasizing the need for a high exposure, typically ≥ 15 mg/kg per day (trimethoprim component) [1-3]. A lowering of the R-breakpoint is planned for 2025 as a result of having reviewed and lowered the ECOFF from 4 to 2 mg/L.

In patients where trimethoprim-sulfamethoxazole is not a suitable agent for treatment because of resistance or, more commonly, sulphonamide intolerance in the patient, selection of therapy is problematic. Various antimicrobials have been used in combinations in *in vitro* studies, but optimal combinations have yet to be identified pending prospective clinical studies [22].

In 2022, the Infectious Diseases Society of America suggested 3 possible approaches for moderate to severe infections [24]. However, these recommendations came at a time when only the first of the meta-analyses listed in Table 3 was available and before most of the recent PK/PD studies noted above. IDSA recommendations have been recently updated [25].

Possible future alternatives

Cefiderocol has good *in vitro* activity and has been used to treat serious infections caused by *S. maltophilia*, although the published numbers are low. In the pivotal studies for registration there were only five cases in the CREDIBLE-CR study [26] and one in the APEKS-NP study [27]. There are case reports of successful treatment of *S. maltophilia* pneumonia with a cefiderocol-based regimen [28,29]. A recent rabbit pneumonia model study showed superior efficacy for cefiderocol compared to trimethoprim-sulfamethoxazole [30]. However, clinical experience with this agent is currently very limited.

Antimicrobial susceptibility testing

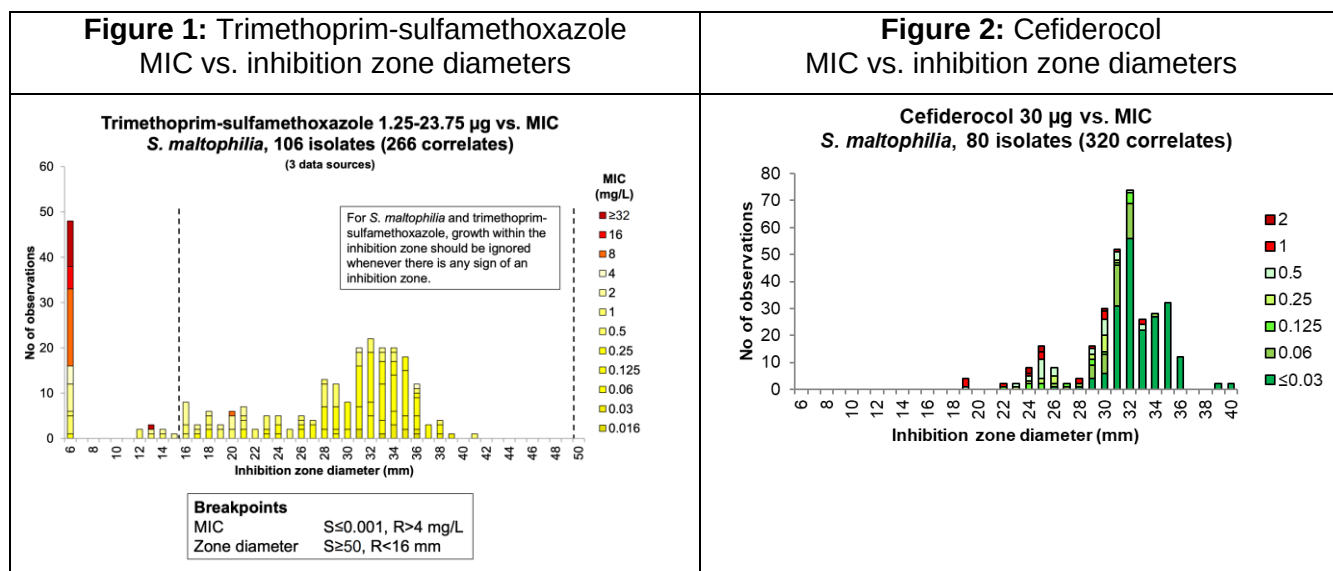
Antimicrobial susceptibility testing of *S. maltophilia* is difficult as results vary significantly with variations in incubation temperature, culture medium and technique [31-39]. Susceptibility test results for agents other than trimethoprim-sulfamethoxazole should be treated with caution as there are insufficient data to support a relationship between susceptibility testing results and clinical outcome with *S. maltophilia* infection.

Susceptibility testing of **trimethoprim-sulfamethoxazole** (Figure 1) can be undertaken using disk diffusion methods. However, reading zone edges is challenging and requires guidance. EUCAST has published guidance in the form of pictures in the breakpoints table. Overcalling resistance is more common than the opposite.

Susceptibility testing of **cefiderocol** (Figure 2) has been less complicated than expected. The difficulty lies in knowing whether wild type *S. maltophilia* isolates are good targets for the agent, and if so, (how close to the ECOFF (0.125 mg/L) should the clinical breakpoint be set. More data are needed before this can be determined. A tentative disk diffusion ECOFF for distinguishing between isolates with and without resistance mechanisms to cefiderocol will be available in the version 15.0 (2025) breakpoint table.

Susceptibility testing of **aztreonam-avibactam** is technically without complications but clinical evidence for a breakpoint is lacking, as is the disk diffusion data needed to define zone diameter correlates.

EUCAST has systematically tested *S. maltophilia* against trimethoprim-sulfamethoxazole and cefiderocol with the following results (Figure 1 and 2).



Conclusions and guidance

There is only weak evidence to support treatment alternatives in *S. maltophilia* infections. Historically trimethoprim-sulfamethoxazole has been regarded the drug of choice, but retrospective clinical data question whether this agent is preferable to other agents.

Alternatives which have been tried and discussed are fluoroquinolones (mainly levofloxacin and moxifloxacin), intravenous minocycline and cefiderocol, but there is scant or no clinical evidence for the usefulness of either of these or for a correlation between susceptibility test results and clinical outcome. Tigecycline may also be an option where intravenous minocycline is unavailable. The role of fluoroquinolones is challenging to assess due to unfavourable PK/PD despite meta-analyses and other studies suggesting superiority to trimethoprim-sulfamethoxazole.

Currently, cefiderocol is the agent with the most favourable *in vitro* activity and may be preferable to the traditional trimethoprim-sulfamethoxazole as first line therapy. However, uncertainty will persist until there are clinical studies to support this conclusion [40]. In summary, the efficacy of all agents and combinations is uncertain and treating physicians should be vigilant in case therapeutic responses are unconvincing.

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